

Polygraph: Reliable ViT Error Detection from Topological Evidence-Flow

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Abstract

Detecting when an image classifier fails is a fundamental challenge for the safe deployment of Vision Transformers (ViTs). Standard error detection mechanisms often depend primarily on the network’s final output logits, largely ignoring the complex token-to-token routing dynamics that construct the prediction. Drawing inspiration from hallucination detection in large language models, we hypothesize that ViTs exhibit recurring topological signatures when forming incorrect predictions. In this work, we introduce Polygraph, a framework that models a frozen ViT’s attention mechanism as a directed graph to capture these latent failure patterns. By deploying an edge-gated MPNN over the final transformer layer, we extract structural evidence-flow features within a single forward pass. These internal signals are then fused with the full-logit vector via a mixture gate. Evaluated on CIFAR-100-C, the graph model provides complementary predictive signal to strong output- and representation-based baselines, while topology ablations indicate a modest additional benefit from explicit graph structure. The combined Polygraph detector achieves the strongest error-detection performance among methods evaluated in the original benchmark and retains its advantage on corruption families entirely unseen during detector training. In exploratory follow-ups, LogitDynamics outperforms our graph ensemble; graph- and set-score fusion nevertheless yield nearly identical observed improvements.

1 Introduction

Vision Transformers (ViTs) (Dosovitskiy et al., 2021) route information among patch tokens, yet reliability is commonly estimated from output scores. Similar output confidence can accompany different internal representations and routing patterns.

To address this, we investigate error prediction through the attributed attention graph induced by a ViT. Given a frozen classifier f and an input

image x with ground-truth label y , let $\hat{y}(x) = \arg \max_c f_c(x)$, and $e(x) = 1[\hat{y}(x) \neq y]$. Our objective is to predict $e(x)$ using only signals available from a single forward pass of the frozen classifier, without external verification or additional classifier evaluations.

While recent approaches such as LogitDynamics (Beigelman and Freiman, 2026) track the layer-wise trajectory of class evidence, they do not explicitly model the token-to-token connectivity underlying these predictions. We propose Polygraph, a method that represents the final ViT block as a directed attributed graph to capture latent routing signatures. Node features are constructed from token hidden states, while edge features encode evidence-flow statistics derived from attention and class signals. An edge-gated MPNN produces a graph-based internal failure score, which is combined with a strong full-logit detector using a learned per-example mixture gate fitted on held-out validation data. The resulting detector improves over either source of evidence alone and retains its advantage on corruption families excluded from detector training. Non-graph controls recover much of the observed performance. In distinct exploratory protocols, we test competitiveness and complementarity with LogitDynamics: graph-score fusion improves LD, but a matched set-score combination gives nearly identical observed performance. This supports useful internal-feature scores without establishing that the follow-up gain requires graph processing.

2 Related Work

ViT Reliability and Attention-Based Diagnostics. A standard baseline for detecting classifier failures is the maximum softmax probability of the predicted class (Hendrycks and Gimpel, 2017). In Vision Transformers, attention and intermediate representations have been exploited for several pur-

poses beyond classification: Chefer et al. (2021) for explanation, Wang et al. (2024) for token pruning, and Yang et al. (2025) for weakly supervised segmentation. Cultrera et al. (2023) use a convolutional autoencoder over ViT attention heatmaps for out-of-distribution detection, but operate on attention heatmaps rather than a token-interaction graph. The closest task-level baseline is LogitDynamics (Beigelman and Freiman, 2026), which predicts ViT classification errors from layer-wise class-logit trajectories and stability statistics derived from intermediate representations, thus reduces each layer to a class-level evidence rather than retaining pairwise interactions among patch tokens.

Graph-Based Error Detection. Natural language processing provides the closest precedents for our work. Earlier approaches construct hand-crafted structural descriptors, such as Kushnareva et al. (2021) and Binkowski et al. (2025). HaluZig (Samaga et al., 2026) models the evolution of attention-graph topology across layers using zigzag persistence. HaluGNN (Kong et al., 2026b) uses hidden states as node features and attention values as weighted edges, while CHARM (Frasca et al., 2026) represents attention and activations as attributes of a token graph and applies edge-aware message passing. CausalGaze (Kong et al., 2026a) further refines attention graphs using gradient-guided counterfactual sensitivity of attention edges. To our knowledge, no prior work applies graph-based message passing to per-image ViT patch-attention graphs for predicting visual classification failures. Polygraph studies this vision-specific setting.

3 Methodology

3.1 Final-Block Attention Graph

For an image x , let the frozen ViT produce class logits $\ell = f(x) \in \mathbb{R}^C$. We construct a graph from its final Transformer block, containing N tokens and H attention heads. Let $\mathcal{V} = \{0, \dots, N-1\}$ denote the token set, with token 0 representing CLS. For head h , $A_{ij}^{(h)}$ denotes the attention assigned by query token i to key token j . We orient the corresponding edge as $j \rightarrow i$.

We retain an edge whenever it receives sufficiently high attention in at least one head:

$$\mathcal{E} = \left\{ (j, i) : j \neq i, \max_h A_{ij}^{(h)} > \tau \right\} \quad (1)$$

Self-edges are excluded from $\mathcal{E}$, but their diagonal attention values are retained as node attributes.

For each token i , the node feature $\mathbf{x}_i$ concatenates its normalized patch coordinates, a CLS indicator, the diagonal attention values $\{A_{ii}^{(h)}\}_{h=1}^H$, and its final-block hidden representation.

3.2 Evidence-Flow Edge Features

For source value $\mathbf{v}_j^{(h)}$, define its head-output projection $\mathbf{u}_j^{(h)} = \mathbf{W}_O^{(h)} \mathbf{v}_j^{(h)}$. Let $\hat{c}, \tilde{c}$ be the highest- and second-highest-logit classes and $\mathbf{w}_c$ their frozen classifier weights. The normalized direction $\mathbf{d} = (\mathbf{w}_{\hat{c}} - \mathbf{w}_{\tilde{c}}) / \|\mathbf{w}_{\hat{c}} - \mathbf{w}_{\tilde{c}}\|_2$ gives predicted-versus-runner-up alignment, not causal attribution.

For each retained edge $j \rightarrow i$ and head h , we then compute an attention-weighted magnitude feature and a signed class-direction feature:

$$\begin{aligned} m_{ji}^{(h)} &= \log \left(1 + A_{ij}^{(h)} \left\| \mathbf{u}_j^{(h)} \right\|_2 \right) \\ c_{ji}^{(h)} &= \text{asinh} \left(A_{ij}^{(h)} \mathbf{d}^\top \mathbf{u}_j^{(h)} \right) \end{aligned} \quad (2)$$

Edges concatenate attention, magnitude and signed alignment across heads; no ground-truth label is used.

3.3 Edge-Gated Graph Detector

We process the attributed graph using a two-layer edge-gated MPNN. Node and edge attributes are first mapped by learned projections to a shared latent representation.

At message-passing layer r , each edge $j \rightarrow i$ produces both a candidate message and an element-wise gate:

$$\begin{aligned} \mu_{ji}^{(r)} &= M_r \left([\mathbf{h}_j^{(r-1)} \parallel \tilde{\mathbf{e}}_{ji}] \right) \\ \gamma_{ji}^{(r)} &= \sigma \left(G_r \left([\mathbf{h}_j^{(r-1)} \parallel \mathbf{h}_i^{(r-1)} \parallel \tilde{\mathbf{e}}_{ji}] \right) \right) \end{aligned} \quad (3)$$

where $\mathbf{h}_i^{(r)}$ is the latent state of node i after layer r , $\tilde{\mathbf{e}}_{ji}$ is the projected edge attribute and M_r and G_r are learned affine maps.

For the incoming neighborhood $\mathcal{N}_i^- = \{j : (j, i) \in \mathcal{E}\}$, the gated messages are aggregated as

$$\bar{\mu}_i^{(r)} = \frac{\sum_{j \in \mathcal{N}_i^-} \gamma_{ji}^{(r)} \odot \mu_{ji}^{(r)}}{\sum_{j \in \mathcal{N}_i^-} \gamma_{ji}^{(r)}} \quad (4)$$

where $\odot$ denotes element-wise multiplication.

Each node is updated from its previous state, the aggregated message, and its log-in-degree

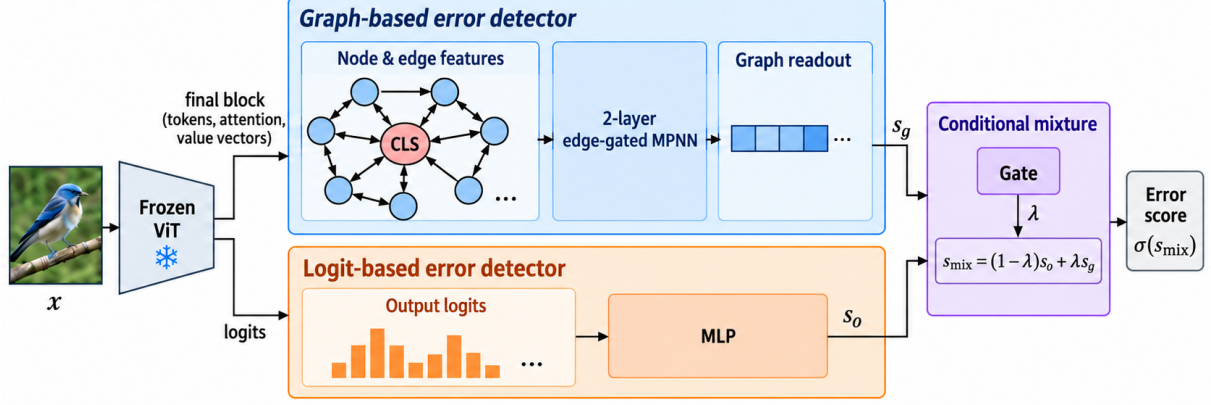


Figure 1: **Original Polygraph architecture.** A final-block edge-gated graph detector and a full-logit MLP are fused by a conditional gate. Later LD score combinations use a separate protocol and logistic combiner.

$\log(1 + |\mathcal{N}_i^-|)$, using an MLP with dropout, a residual connection, LayerNorm, and ReLU.

After message passing, we combine the final CLS-node state with a learned weighted pooling of all token states. Let ψ be a learned scalar node scorer and ρ the final graph decoder. The graph error logit is

$$\beta_i = \frac{\exp(\psi(\mathbf{h}_i))}{\sum_{j \in \mathcal{V}} \exp(\psi(\mathbf{h}_j))} \quad (5)$$

$$s_g(x) = \rho \left([\mathbf{h}_0 \parallel \sum_{i \in \mathcal{V}} \beta_i \mathbf{h}_i] \right)$$

3.4 Output Expert and Conditional Fusion

The output expert is an MLP on the full logit vector, standardized per class on detector-training data. Both experts are trained independently with class-weighted binary cross-entropy against the error label $e(x)$ defined above.

Once trained, both experts are frozen. Their scores are independently calibrated on held-out meta-validation data, producing calibrated logits $z_o(x)$ and $z_g(x)$. A small gate receives the standardized pair of expert scores and predicts an example-specific graph weight $\lambda(x) \in [0, 1]$:

$$z_{\text{mix}}(x) = (1 - \lambda(x)) z_o(x) + \lambda(x) z_g(x) \quad (6)$$

Equation (6) is applied separately in each of five grouped meta-validation folds. Each fold fits its score standardizer, scalar logistic calibrators, and gate on four folds, using the remaining fold for gate early stopping. At test time, we average the five mixture logits; both frozen experts are shared across folds.

All calibration and gate fitting use disjoint meta-validation photographs.

Implementation details. In our experiments, the ViT has $N = 197$ tokens, $H = 12$ heads, $D = 768$, and per-head dimension 64. We use $\tau = 0.02$, resulting in approximately 7,600 directed edges per image. Node features contain 16 base attributes plus the 768-dimensional hidden state, and edge attributes contain 36 values (12 attention, 12 magnitude, and 12 class-direction features). The graph detector uses latent width 64, two message-passing layers, and dropout 0.15. The output expert has dimensions $100 \rightarrow 64 \rightarrow 32 \rightarrow 1$, and the mixture gate uses $2 \rightarrow 16 \rightarrow 1$. We use five grouped folds for mixture fitting.

4 Experimental Setup

This section describes the original Polygraph experiments; the follow-ups use the distinct protocols in Section 6.

4.1 Datasets and Splits

We evaluate on CIFAR-100 and CIFAR-100-C using a frozen ViT-B/16 pretrained on ImageNet-21k and fine-tuned on CIFAR-100. We construct an error-balanced benchmark by sampling equal numbers of correct and incorrect predictions within each source-severity stratum. All versions of the same original image remain in one train/validation/test partition. We consider the main benchmark and a weather-family holdout in which selected corruption families are excluded from detector training and used to evaluate distribution-shift generalization. Model selection uses only the corresponding validation split, and results for these original experiments are on held-out test data.

4.2 Training and Baselines

We select models by validation AUROC with early stopping and repeat all learned detectors over three random seeds. We compare against maximum softmax probability (MSP) and a strict output-only MLP trained on the complete classifier-logit vector.

5 Results

5.1 Main Error Detection

Table 1 compares detectors on the main CIFAR-100/CIFAR-100-C test split. Polygraph improves mean AUROC over the strict full-logit MLP, while also improving AUPRC and AURC. The standalone graph expert provides a smaller mean AUROC gain. Throughout this section, *graph expert* denotes the selected two-layer edge-gated MPNN, and *Polygraph* denotes its conditional fusion with the output MLP.

Method	AUROC $\uparrow$	AUPRC $\uparrow$	AURC $\downarrow$
MSP	0.8651	0.8345	0.2218
Output MLP	0.8794	0.8583	0.2158
Graph expert	0.8883	0.8682	0.2105
Polygraph	0.8939	0.8748	0.2071

Table 1: Main-test performance, averaged over three seeds for learned detectors. MSP is deterministic. Bold marks the best result in each column.

Paired photo-group bootstrap (2,000 draws) gives positive 95% intervals for Polygraph–output AUROC in each seed. At a 50% flagging budget per detector, the graph expert detects 37.4% of errors missed by the output MLP (seed mean), supporting fusion despite less certain standalone AUROC gains.

Polygraph has the highest mean AUROC at all five corruption severities; the standalone graph expert also exceeds the output MLP at each severity.

5.2 Held-Out Corruption Generalization

Under the weather-family holdout, conditional Polygraph exceeds the output MLP on both the full test and unseen-corruption subset (Table 2). On the unseen subset, it also exceeds MSP, unlike either standalone expert. This is corruption-family generalization within the original benchmark.

5.3 Representation and Topology Ablations

Feature ablations. In the single-seed TransformerConv validation study, adding class direc-

Method	Full test AUROC $\uparrow$	Unseen only AUROC $\uparrow$
MSP	0.8625	0.8844
Output MLP	0.8690	0.8816
Graph expert	0.8764	0.8840
Polygraph	0.8820	0.8912

Table 2: Weather-holdout AUROC, averaged over three seeds for learned detectors. Full test includes seen and unseen corruption families; Unseen only contains the held-out weather and extra corruptions.

tion to base-node/raw-attention inputs raised AUROC from 0.8348 to 0.8660; magnitude alone gave 0.8369, and both gave 0.8655. With hidden-node inputs, full edge features scored 0.8863 versus 0.8754 for raw attention.

Non-graph controls. Table 3 compares models under disjoint validation data for expert selection and fusion. All models receive the same hidden-node inputs. The node/edge-set control pools node and edge features separately, without retaining their incidence; the endpoint-set control retains local source–target–edge records but performs no message passing. Both nearly match TransformerConv. The edge-gated MPNN has the highest mean AUROC.

Internal detector	AUROC $\uparrow$
TransformerConv	0.8845
Token set, no edges [†]	0.8838
Node/edge sets, no incidence	0.8834
Endpoint sets, no message passing	0.8813
Edge-gated MPNN	0.8883

Table 3: Main-test topology controls under the strict validation protocol. All models use hidden-node inputs. Values are three-seed means except [†], which uses seed 7 only.

A separate five-seed rich-feature study found graph–set AUROC +0.0062 (95% interval [0.0026, 0.0098]); the attention-only contrast was −0.0037 ([−0.0083, 0.0010]). Its cohort, features and budget differ from the later raw-attention ensembles, so these are not controlled replications.

Architecture selection. Validation screening selected the two-layer edge-gated MPNN over tested TransformerConv, GINE and GATv2 operators and one-/four-message-passing-layer variants. Joint representations of the final four ViT blocks did not improve the tested models over the final-block expert.

6 Controlled Follow-up Studies

These exploratory studies reuse 800 previously examined development photographs, each with nine views: clean, plus Gaussian noise, motion blur, fog, and JPEG at severities 3/5. Unlike the original benchmark, this cohort is not error-balanced. We average seed-specific AUROCs over seeds 7/17/27; these are not independent confirmation.

Models and protocols. G averages four separately trained two-layer edge-gated detectors at attention layers 3/6/9/12; S uses similarly sized set detectors. Within each layer, both arms share H12-based 784-dimensional node values and 12 attention-edge values (zeroed at or below 0.02). Sets pool nodes/edges without incidence. Existing fits used 1,600/400 training/selection photographs and 20 epochs. Unlike the joint multi-block models above, these combine scores: G reaches 0.8889 versus 0.8831 for one final-layer detector, without isolating layer diversity from generic ensembling.

The fixed ViT-B LogitDynamics adaptation (D) (Beigelman and Freiman, 2026) uses auxiliary linear CLS class heads and a linear error readout. Its allocation is 1,200/800/400/800 photographs for head fitting/readout fitting/checkpoint selection/evaluation. Heads train for 16 epochs; readouts for 100, selected by validation AP. On all 800 evaluation photographs, D reaches 0.8981 versus 0.8889 for G ; $G - D = -0.00917$ (95% interval $[-0.01587, -0.00236]$). Inputs, supervision and budgets differ; this does not compare D with the original conditional Polygraph.

Complementarity. A fixed 400/400 subdivision fits and assesses standardized L2 logistic combinations of preserved scores; scaling and fitting use only the first half. These are not the original non-linear mixture gate. G/S are mean probabilities; D is its uncalibrated error logit. D' uses another LD seed with cyclic, overlapping pairings. Table 4 shows that weaker standalone scores nevertheless improve LD: $DG - D = +0.007203$, adjusted interval $[0.003415, 0.011330]$. The graph-set fusion difference is only $+0.000043$, interval $[-0.001311, 0.001508]$: no graph-specific advantage is demonstrated, nor is equivalence. The secondary $DG - DD'$ difference is $+0.005213$ (95% $[0.001942, 0.008673]$), favoring graph fusion over this two-seed LD control, not generic ensembling generally; costs are unmatched.

Detector	AUROC
LD D	0.9001
Graph ensemble G	0.8956
Set ensemble S	0.8950
$D + G$	0.9073
$D + S$	0.9073
$D + D'$	0.9021

Table 4: Mean seed-specific AUROC on 400 fusion-assessment photographs. Do not compare with the separate 800-photo evaluation or original benchmark.

LD decomposition. Reusing the auxiliary heads, three readouts are fitted with the original recipe. Six original-classifier features (A) yield 0.8636; adding the final-layer auxiliary block (B), 0.8948; all 12 auxiliary numeric blocks plus A (C), 0.8990; adding dynamics (preserved D), 0.8981. Each head supplies its logit for the original classifier’s predicted class and its five largest logits for other classes. A is not the full-logit MLP. $C - B = +0.004135$, adjusted interval $[0.000428, 0.007911]$, supports a small intermediate-layer contribution. The largest observed increment is $B - A$; dynamics, including class-identity features, did not improve this recipe. These ablations do not causally explain the LD-GNN gap.

Uncertainty. The new studies use 2,000 paired photo-group bootstrap draws, sharing views and draws across methods/seeds. Fusion resamples fitting and assessment photographs and refits combiners; decomposition resamples evaluation only. Bonferroni adjustment over $DG - D$, $DG - DS$, and $C - B$ gives individual $98\frac{1}{3}\%$ percentile intervals. Fusion intervals condition on base detectors and the partition; decomposition intervals on all fitted readouts. Neither corrects prior development exposure.

7 Conclusion

Polygraph combines a graph-based internal detector with a full-logit expert, improving original-benchmark error detection and retaining its benefit under corruption-family holdout. Exploratory follow-ups find complementary value from graph/set scores alongside LD, without demonstrating a graph-specific fusion advantage. Intermediate auxiliary-head projections also improve the tested LD readout. These findings support useful internal signals without establishing topology’s necessity.

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